EXTRA PROBLEMS · 1ST MIDTERM · CONCEPTS

1st Midterm: Short Conceptual Questions

Short conceptual questions on Units 1 to 5, worth 2 points each, in the style of the true/false questions of the final exams. Written for this portal and solved step by step; they are practice material, not official exercises.

8 short conceptual questions on Units 1 to 5, in the style of the true/false questions of the final exams. Each is worth 2 points and asks for a justified answer. They are practice for the reasoning behind the Midterm 1 problem; the exam itself is one problem of 10 points, so these questions do not appear in the Exam simulator.

Conceptual question · Unit 1 · 2 points · 15 min · Conceptual / fast

C01 — Intrinsic acceleration on a parabolic helix: constant angular rate is not constant speed

A point particle P moves with respect to the inertial frame $S_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ along the curve given in cylindrical coordinates (R, φ, z), with basis $\{\mathbf{e}_R, \mathbf{e}_\varphi, \mathbf{e}_z\}$, by

$$R = a, \qquad z = \frac{a}{2}\,\varphi^2, \qquad \varphi \geq 0,$$

with $a > 0$ a constant length: a helix wound on the cylinder $R = a$ whose pitch grows with φ. Decide whether each statement is TRUE or FALSE and justify the answer. A verdict without a justification scores nothing.

Part (a) · 1 point

Constant angular rate

The particle moves with the time law $\varphi = \omega t$, $\omega > 0$ constant. Statement: "Since $\dot\varphi$ is constant, the tangential acceleration of P is zero, and at the point $\varphi = 1$ the acceleration of P is purely normal."

Step 1. Velocity in cylindrical coordinates

With $\mathbf{r} = R\,\mathbf{e}_R + z\,\mathbf{e}_z$, $\dot{\mathbf{e}}_R = \dot\varphi\,\mathbf{e}_\varphi$ and $\dot{\mathbf{e}}_\varphi = -\dot\varphi\,\mathbf{e}_R$ (Notes §2.6), the velocity is $\mathbf{v} = \dot R\,\mathbf{e}_R + R\dot\varphi\,\mathbf{e}_\varphi + \dot z\,\mathbf{e}_z$. Here $\dot R = 0$, $\dot\varphi = \omega$ and, by the chain rule, $\dot z = \frac{dz}{d\varphi}\dot\varphi = a\varphi\,\omega$:

$$\mathbf{v} = a\omega\,\mathbf{e}_\varphi + a\omega\varphi\,\mathbf{e}_z, \qquad |\mathbf{v}| = a\omega\sqrt{1+\varphi^2}$$

The speed is not constant, so the statement already fails: a constant angular rate does not mean a constant speed when the curve climbs faster and faster.

Step 2. Acceleration

$\mathbf{a} = (\ddot{R} - R\dot{\varphi}^2)\,\mathbf{e}_R + (R\ddot{\varphi} + 2\dot{R}\dot{\varphi})\,\mathbf{e}_\varphi + \ddot{z}\,\mathbf{e}_z$, with $\ddot{\varphi} = 0$ and $\ddot{z} = \frac{d}{dt}(a\omega\varphi) = a\omega\dot{\varphi} = a\omega^2$:

$$\mathbf{a} = -a\omega^2\,\mathbf{e}_R + a\omega^2\,\mathbf{e}_z, \qquad |\mathbf{a}|^2 = 2a^2\omega^4$$

Step 3. Tangential and normal components

The tangential acceleration is $a_t = \frac{d|\mathbf{v}|}{dt}$ (Notes §2.5). With the chain rule $\frac{d}{dt}\sqrt{1+\varphi^2} = \frac{\varphi\,\dot{\varphi}}{\sqrt{1+\varphi^2}}$:

$$a_t = a\omega\,\frac{\varphi\,\omega}{\sqrt{1+\varphi^2}} = \frac{a\omega^2\varphi}{\sqrt{1+\varphi^2}}$$

The same result follows from $a_t = \mathbf{a}\cdot\mathbf{v}/|\mathbf{v}| = a^2\omega^3\varphi/(a\omega\sqrt{1+\varphi^2})$. At $\varphi = 1$:

$$a_t = \frac{a\omega^2}{\sqrt{2}} \neq 0, \qquad a_n = \sqrt{|\mathbf{a}|^2 - a_t^2} = a\omega^2\sqrt{2 - \tfrac{1}{2}} = \sqrt{\tfrac{3}{2}}\,a\omega^2$$

The acceleration has a tangential part, so the statement is FALSE. For later use: at $\varphi = 1$, $\mathbf{e}_t = (\mathbf{e}_\varphi + \mathbf{e}_z)/\sqrt{2}$, $a_t\mathbf{e}_t = \frac{1}{2}a\omega^2(\mathbf{e}_\varphi + \mathbf{e}_z)$ and

$$\mathbf{a}_n = \mathbf{a} - a_t\mathbf{e}_t = a\omega^2\big(-\mathbf{e}_R - \tfrac{1}{2}\mathbf{e}_\varphi + \tfrac{1}{2}\mathbf{e}_z\big), \qquad \mathbf{e}_n = \frac{-2\,\mathbf{e}_R - \mathbf{e}_\varphi + \mathbf{e}_z}{\sqrt{6}}$$

KEY POINTS

- FALSE.
- $\mathbf{v} = a\omega\,\mathbf{e}_\varphi + a\omega\varphi\,\mathbf{e}_z$, $|\mathbf{v}| = a\omega\sqrt{1+\varphi^2}$ grows with time; $\mathbf{a} = a\omega^2(-\mathbf{e}_R + \mathbf{e}_z)$.
- $a_t = a\omega^2\varphi/\sqrt{1+\varphi^2}$: at $\varphi = 1$, $a_t = a\omega^2/\sqrt{2} \neq 0$ and $a_n = \sqrt{3/2}\,a\omega^2$.

Part (b) · 1 point

Constant speed

The same curve is now traversed at a constant speed v_0. Statement: "At the point $\varphi = 1$ the acceleration of P is perpendicular to its velocity, has magnitude $\sqrt{6}\,v_0^2/(4a)$ and points along the same unit vector $\mathbf{e}_n$ as the normal acceleration of part (a)."

Step 1. Intrinsic acceleration

In the intrinsic basis (Notes §2.5), $\mathbf{a} = \ddot{s}\,\mathbf{e}_t + \dfrac{\dot{s}^2}{\rho}\,\mathbf{e}_n$. With $\dot{s} = v_0$ constant, $\ddot{s} = 0$, so $\mathbf{a} = \dfrac{v_0^2}{\rho}\,\mathbf{e}_n$, which is perpendicular to $\mathbf{v} = v_0\,\mathbf{e}_t$. It is not zero: the trajectory is curved.

Step 2. Curvature at $\varphi = 1$

The curvature depends only on the geometry. From part (a) at $\varphi = 1$, $\kappa = \dfrac{a_n}{|\mathbf{v}|^2} = \dfrac{\sqrt{3/2}\,a\omega^2}{2a^2\omega^2} = \dfrac{\sqrt{6}}{4a}$.
Independent check with the parameter φ: $\mathbf{r}' = a(-\sin\varphi, \cos\varphi, \varphi)$ and $\mathbf{r}'' = a(-\cos\varphi, -\sin\varphi, 1)$ in $\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$, so

$$\mathbf{r}' \times \mathbf{r}'' = a^2\big(\cos\varphi + \varphi\sin\varphi,\ \sin\varphi - \varphi\cos\varphi,\ 1\big), \qquad |\mathbf{r}' \times \mathbf{r}''| = a^2\sqrt{2+\varphi^2}, \qquad |\mathbf{r}'| = a\sqrt{1+\varphi^2}$$

$$\kappa = \frac{|\mathbf{r}' \times \mathbf{r}''|}{|\mathbf{r}'|^3} = \frac{\sqrt{2+\varphi^2}}{a\,(1+\varphi^2)^{3/2}}, \qquad \kappa(1) = \frac{\sqrt{3}}{a\,2^{3/2}} = \frac{\sqrt{6}}{4a}$$

Step 3. Magnitude and direction

$$|\mathbf{a}| = \kappa\,v_0^2 = \frac{\sqrt{6}\,v_0^2}{4a}$$

The principal normal is $\mathbf{e}_n = \dfrac{d\mathbf{e}_t/ds}{|d\mathbf{e}_t/ds|}$, a function of the point of the curve only (Notes §2.4), so it is the same vector $(-2\,\mathbf{e}_R - \mathbf{e}_\varphi + \mathbf{e}_z)/\sqrt{6}$ found in (a). The three claims hold: TRUE.

KEY POINTS

- TRUE.
- $\mathbf{a} = \dot{v}\,\mathbf{e}_t + \dfrac{v^2}{\rho}\,\mathbf{e}_n$ with $\dot{v} = 0$: $\mathbf{a} \perp \mathbf{v}$.
- $\kappa(\varphi) = \dfrac{\sqrt{2+\varphi^2}}{a(1+\varphi^2)^{3/2}}$, $\kappa(1) = \dfrac{\sqrt{6}}{4a}$, $|\mathbf{a}| = \kappa v_0^2 = \dfrac{\sqrt{6}\,v_0^2}{4a}$; $\mathbf{e}_n$ depends only on the curve.

Interpretation and checks

- **Dimensional check:** $a\omega^2$ is m s^{-2}; κ is m^{-1} and κv_0^2 is m s^{-2}.
- **Limits:** at $\varphi = 0$ the tangent is horizontal, but $\kappa(0) = \sqrt{2}/a$ is larger than the $1/a$ of the circle $R = a$ because the curve also bends upwards ($d^2z/d\varphi^2 = a \neq 0$); for $\varphi \to \infty$, $\kappa \approx 1/(a\varphi^2) \to 0$: the turns become almost vertical lines.
- **Common mistakes in the exam:** (1) confusing a constant angular rate $\dot{\varphi}$ with a constant speed; (2) saying "constant speed implies zero acceleration"; (3) believing that $\mathbf{e}_t$, $\mathbf{e}_n$ or ρ depend on how fast the curve is traversed.

`Conceptual question · Unit 2 · 2 points · 15 min · Conceptual / fast`

CO2 — Motion in a cubic potential: a separatrix that is never crossed and an asymmetric well

A point particle P of mass m moves along the x axis of an inertial frame under the single conservative force derived from the potential energy

$$V(x) = V_0 \left[\left(\frac{x}{b} \right)^3 - 3\,\frac{x}{b} \right],$$

with $V_0 > 0$ (an energy) and $b > 0$ (a length) constants. The figure shows $V(x)$ and two energy levels. Decide whether each statement is TRUE or FALSE and justify the answer. A verdict without a justification scores nothing.

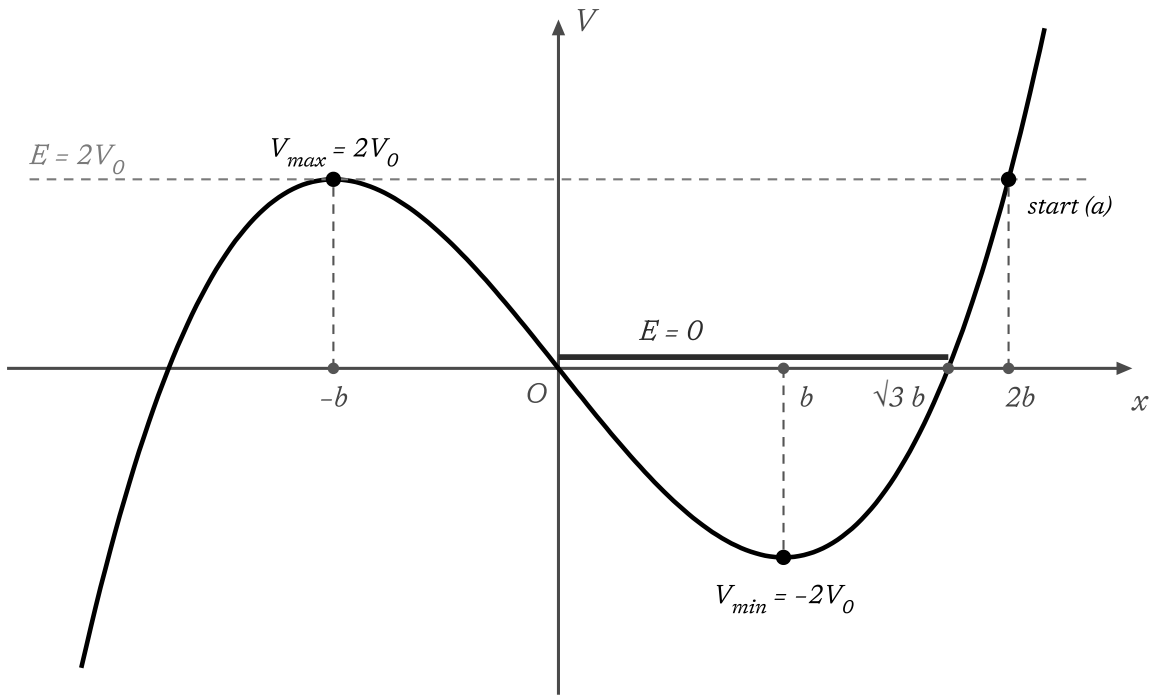


Figure 1. Potential $V(x) = V_0\left[(x/b)^3 - 3x/b\right]$ with the energy levels $E = 2V_0$ (released at $x = 2b$) and $E = 0$ (released at $x = 0$).

Part (a) · 1 point

Released at $x = 2b$

Statement: "If P is released from rest at $x = 2b$, it moves to the left, reaches the top of the barrier at $x = -b$ after a finite time and then escapes towards $x \to -\infty$."

Step 1. Energy and direction of the motion

Write $u = x/b$, so $V = V_0(u^3 - 3u)$. By the chain rule $\frac{dV}{dx} = \frac{dV}{du}\frac{du}{dx} = \frac{3V_0}{b}(u^2 - 1)$: the equilibria are $x = \pm b$, with $V(-b) = 2V_0$ (maximum, $V'' < 0$) and $V(b) = -2V_0$ (minimum). Released from rest at $x = 2b$, the energy is $E = V(2b) = V_0(8 - 6) = 2V_0$ and the initial force is $F = -\frac{dV}{dx}\big|_{2b} = -\frac{3V_0}{b}(4 - 1) = -\frac{9V_0}{b} < 0$: P moves to the left.

Step 2. The kinetic energy factorises

Energy conservation (Notes §4.5) gives $\frac{1}{2}m\dot{x}^2 = E - V$, and

$$E - V = V_0\big(2 - u^3 + 3u\big) = V_0\,(u+1)^2\,(2-u)$$

(check: $(u+1)^2(2-u) = (u^2 + 2u + 1)(2 - u) = -u^3 + 3u + 2$). For $-1 < u < 2$ it is positive, and it vanishes at $u = -1$ with a **double** root.

Step 3. Time to approach $x = -b$

On the way to the left $\dot{x} = -\sqrt{2V_0/m}\,(u+1)\sqrt{2-u}$, so $dt = \dfrac{-b\,du}{\sqrt{2V_0/m}\,(u+1)\sqrt{2-u}}$. Substitute $w = \sqrt{2-u}$: $u = 2 - w^2$, $du = -2w\,dw$, $u + 1 = 3 - w^2$, with $w = 0$ at the start ($u = 2$) and $w \to \sqrt{3}$ at the barrier ($u \to -1$):

$$t(w) = \frac{2b}{\sqrt{2V_0/m}} \int_0^{w} \frac{dw'}{3 - w'^2}$$

With the antiderivative $\int \frac{dw}{3-w^2} = \frac{1}{2\sqrt{3}} \ln \frac{\sqrt{3}+w}{\sqrt{3}-w}$ and Barrow's rule (the lower limit gives $\ln 1 = 0$):

$$t(w) = b\sqrt{\frac{m}{6V_0}}\;\ln\frac{\sqrt{3}+w}{\sqrt{3}-w} \xrightarrow[w\to\sqrt{3}]{} \infty$$

For instance, P passes $x = 0$ ($w = \sqrt{2}$) at $t = b\sqrt{m/(6V_0)}\,\ln\big(5 + 2\sqrt{6}\big) \approx 0.936\,b\sqrt{m/V_0}$, but it needs an infinite time to reach $x = -b$. The particle tends to the unstable equilibrium and never escapes: the statement is FALSE.

KEY POINTS

- FALSE.
- $V(2b) = 2V_0 = V(-b)$: the energy equals the top of the barrier. $F(2b) = -9V_0/b < 0$, so P does start to the left.
- $E - V = V_0(u+1)^2(2-u)$ with $u = x/b$: $|\dot{x}| \propto |x+b|$ near $x = -b$ and the time $t = b\sqrt{m/(6V_0)}\,\ln\dfrac{\sqrt{3}+w}{\sqrt{3}-w}$, $w = \sqrt{2 - x/b}$, diverges as $x \to -b$: P approaches the barrier top asymptotically and never crosses it.

Part (b) · 1 point

Released at $x = 0$

Statement: "If P is released from rest at $x = 0$, it oscillates about the stable equilibrium $x = b$ between $x = 0$ and $x = 2b$, symmetrically about $x = b$."

Step 1. Energy and initial force

$E = V(0) = 0$. The initial force is $F = -\frac{dV}{dx}\big|_0 = -\frac{3V_0}{b}(0-1) = \frac{3V_0}{b} > 0$: P moves to the right, into the well of the minimum $x = b$.

Step 2. Turning points

They solve $V(x) = E = 0$: $V_0(u^3 - 3u) = V_0\,u\,(u-\sqrt{3})(u+\sqrt{3}) = 0$, so $u \in \{-\sqrt{3}, 0, \sqrt{3}\}$. Between $u = 0$ and $u = \sqrt{3}$ we have $V < 0 = E$ and the kinetic energy is positive; the motion is confined to $0 \le x \le \sqrt{3}\,b$. At both ends $dV/dx \neq 0$ ($-3V_0/b$ at $x = 0$, $+6V_0/b$ at $x = \sqrt{3}\,b$), so they are simple turning points reached in a finite time, and the motion is a periodic oscillation (Notes §4.5).

Step 3. Not symmetric, not up to $2b$

$x = 2b$ is never reached, since $V(2b) = 2V_0 > E$. The interval $[0, \sqrt{3}\,b]$ extends b to the left of the minimum and only $(\sqrt{3}-1)\,b \approx 0.732\,b$ to its right: the cubic well is steeper on the right. Symmetry about the minimum holds only for a parabolic well, i.e. for small oscillations, whose frequency follows from $V''(b) = \frac{1}{b^2}\frac{d^2V}{du^2}\big|_{u=1} = \frac{6V_0}{b^2}$: $\omega^2 = \dfrac{6V_0}{mb^2}$ (Notes §4.6). The statement is FALSE.

KEY POINTS

- FALSE.
- $E = V(0) = 0$; $V = V_0 u(u-\sqrt{3})(u+\sqrt{3})$: turning points $x = 0$ and $x = \sqrt{3}\,b \approx 1.732\,b$ (not $2b$, where $V = 2V_0 > E$).
- $x = b$ is stable ($V'' > 0$), small-oscillation frequency $\omega^2 = 6V_0/(mb^2)$; the motion is periodic but the distances to the turning points are b and $0.732\,b$.

Interpretation and checks

- **Dimensional check:** V_0/b is a force (N); $b\sqrt{m/V_0}$ is $\mathrm{m}\sqrt{\mathrm{kg/J}} = \mathrm{s}$; $\omega^2 = 6V_0/(mb^2)$ is s^{-2}.
- **Limits:** with E slightly below $2V_0$, P would turn back just before $x = -b$ after a long but finite time; slightly above, it would cross the barrier and run to $x \to -\infty$ ($V \to -\infty$, unbounded speed). $E = 2V_0$ is the separatrix between both behaviours.
- **Common mistakes in the exam:** (1) assuming that "enough energy to reach the top" means crossing it; (2) placing the turning points symmetrically about the minimum; (3) forgetting to check the sign of the initial force to know where the particle goes.

Conceptual question · Unit 2 · 2 points · 15 min · Conceptual / fast

C03 — Energy of a pendulum hanging from a slider: one particle, the system, and a driven support

A slider A of mass M moves along a smooth, fixed horizontal rail (the x axis direction of the inertial frame $S_0 = \{O; \mathbf{i}, \mathbf{j}\}$, y upward, gravity $-g\,\mathbf{j}$); its abscissa is x_A. A bob P of mass m hangs from A on a massless, inextensible string of length ℓ that stays taut. The angle θ is measured from the downward vertical, positive towards $+x$, and $\mathbf{e}_r = \sin\theta\,\mathbf{i} - \cos\theta\,\mathbf{j}$ (from A to P), $\mathbf{e}_\theta = \cos\theta\,\mathbf{i} + \sin\theta\,\mathbf{j}$. The motion is planar. Decide whether each statement is TRUE or FALSE and justify the answer. A verdict without a justification scores nothing.

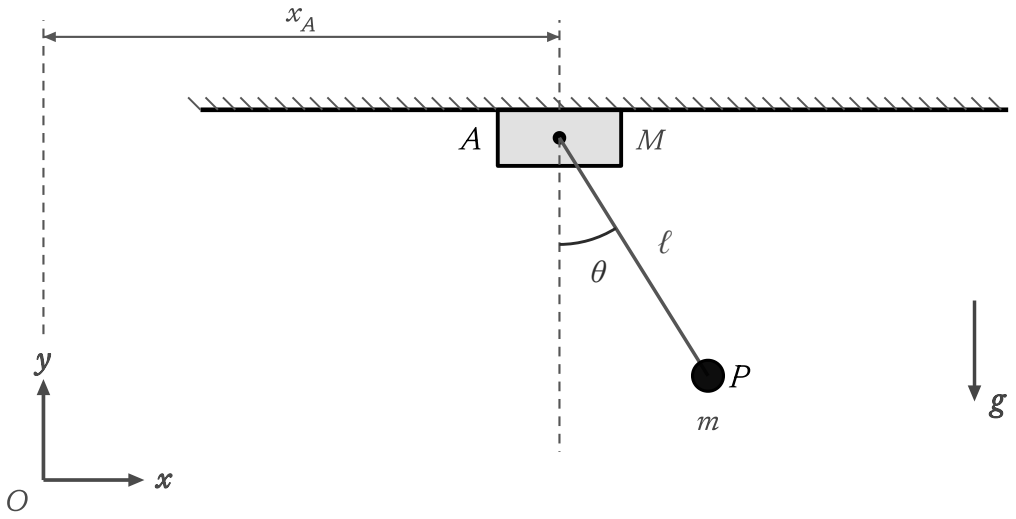


Figure 1. Slider A (mass M) on a smooth horizontal rail, abscissa x_A; bob P (mass m) on a string of length ℓ, angle θ from the downward vertical, positive towards $+x$.

Part (a) · 1 point

Free slider

*The slider moves freely on the rail. Statement: "The mechanical energy of the bob alone, $E_P = \frac{1}{2}m|\mathbf{v}_P|^2 + mg\,y_P$, is **not** conserved, but the mechanical energy of the system slider plus bob, $E = \frac{1}{2}M\dot{x}_A^2 + E_P$, is conserved."*

Step 1. Power of the tension on the bob

The string pulls P towards A: $\mathbf{T}_P = -T\,\mathbf{e}_r, T \geq 0$. The absolute velocity of P is $\mathbf{v}_P = \dot{x}_A\,\mathbf{i} + \ell\dot{\theta}\,\mathbf{e}_\theta$ (derivative of $\mathbf{r}_P = x_A\mathbf{i} + \ell\mathbf{e}_r$ with $\dot{\mathbf{e}}_r = \dot{\theta}\,\mathbf{e}_\theta$). With $\mathbf{e}_r \cdot \mathbf{i} = \sin\theta$ and $\mathbf{e}_r \cdot \mathbf{e}_\theta = 0$:

$$\dot{W}_{T,P} = \mathbf{T}_P \cdot \mathbf{v}_P = -T\dot{x}_A \sin\theta$$

The tension is orthogonal to the velocity of P **relative to** A, but not to its absolute velocity. By the work-energy theorem with the weight as the only conservative force on P (Notes §4.5), $\dot{E}_P = -T\dot{x}_A \sin\theta$, which is not zero whenever the slider moves with the string inclined: E_P is not conserved.

Step 2. The system

On A act $M\mathbf{g}$ and the rail reaction (both vertical, power zero against $\dot{x}_A\mathbf{i}$: smooth rail) and the string force $\mathbf{T}_A = +T\,\mathbf{e}_r$, of power $T\dot{x}_A\sin\theta$. Hence $\frac{d}{dt}\left(\frac{1}{2}M\dot{x}_A^2\right) = T\dot{x}_A\sin\theta$ and

$$\dot{E} = T\dot{x}_A\sin\theta - T\dot{x}_A\sin\theta = T\,\mathbf{e}_r\cdot(\mathbf{v}_A - \mathbf{v}_P) = -T\ell\dot{\theta}\,\mathbf{e}_r\cdot\mathbf{e}_\theta = 0$$

The internal pair of an inextensible string does no net work (Notes §4.4), so E is conserved. The statement is TRUE.

Step 3. How much energy moves from P to A

The horizontal external force is zero, so $p_x = (M+m)\dot{x}_A + m\ell\dot{\theta}\cos\theta$ is conserved, equal to 0 for a release from rest at θ_0. At $\theta = 0$: $\dot{x}_A = -\dfrac{m\ell\dot{\theta}}{M+m}$. At $\theta = 0$, $\mathbf{e}_\theta = \mathbf{i}$, so $\mathbf{v}_P = (\dot{x}_A + \ell\dot{\theta})\,\mathbf{i}$ with $\dot{x}_A + \ell\dot{\theta} = \dfrac{M\ell\dot{\theta}}{M+m}$, and $y_P = -\ell\cos\theta$. Energy conservation between the release (θ_0, at rest) and $\theta = 0$:

$$\frac{1}{2}M\frac{m^2\ell^2\dot{\theta}^2}{(M+m)^2} + \frac{1}{2}m\frac{M^2\ell^2\dot{\theta}^2}{(M+m)^2} - mg\ell = -mg\ell\cos\theta_0$$

The two kinetic terms add up to $\dfrac{Mm(m+M)\ell^2\dot{\theta}^2}{2(M+m)^2} = \dfrac{Mm\,\ell^2\dot{\theta}^2}{2(M+m)}$, so $\dfrac{Mm\,\ell^2\dot{\theta}^2}{2(M+m)} = mg\ell(1-\cos\theta_0)$, which gives $\ell\dot{\theta}^2 = \dfrac{2g(1-\cos\theta_0)(M+m)}{M}$ and

$$\tfrac{1}{2}M\dot{x}_A^2 = \frac{M}{2}\,\frac{m^2\ell^2\dot{\theta}^2}{(M+m)^2} = \frac{m^2 g\ell\,(1-\cos\theta_0)}{M+m} > 0$$

which is exactly what E_P has lost.

KEY POINTS

- TRUE.
- Tension on P: $-T\mathbf{e}_r$, power $-T\mathbf{e}_r \cdot \mathbf{v}_P = -T\dot{x}_A \sin\theta \neq 0$ in general, so $\dot{E}_P = -T\dot{x}_A \sin\theta$.
- Tension on A: $+T\mathbf{e}_r$, power $+T\dot{x}_A \sin\theta$; the pair has total power $T\mathbf{e}_r \cdot (\mathbf{v}_A - \mathbf{v}_P) = 0$ (inextensible string); rail reaction and $M\mathbf{g}$ are vertical, orthogonal to $\dot{x}_A\mathbf{i}$: $\dot{E} = 0$.
- Example: released from rest at θ_0, at $\theta = 0$ the slider has $\frac{1}{2}M\dot{x}_A^2 = \dfrac{m^2 g\ell(1-\cos\theta_0)}{M+m}$, energy that P has lost.

Part (b) · 1 point

Driven slider

A motor now imposes $x_A(t) = b\sin(\Omega t)$. Let S_1 be the frame that translates with A. Statement: "In S_1 the tension does no work on P and the weight is conservative; therefore $E_{rel} = \frac{1}{2}m\ell^2\dot{\theta}^2 - mg\ell\cos\theta$ is a first integral of the motion of the bob."

Step 1. Forces and work in S_1

S_1 translates with acceleration $\ddot{x}_A\mathbf{i}$, so (Notes §4.9) the forces on P in S_1 are $m\mathbf{g}$, $-T\mathbf{e}_r$ and the inertia force $\mathbf{F}_{in} = -m\ddot{x}_A\,\mathbf{i}$ (no Coriolis or centrifugal terms: $\boldsymbol{\omega}_{10} = \mathbf{0}$). With $\mathbf{v}_{rel} = \ell\dot{\theta}\,\mathbf{e}_\theta$ the tension does no work, and the weight has the potential $-mg\ell\cos\theta$. The power of the inertia force is

$$\mathbf{F}_{in}\cdot\mathbf{v}_{rel} = -m\ddot{x}_A\,\ell\dot{\theta}\,(\mathbf{i}\cdot\mathbf{e}_\theta) = -m\ddot{x}_A\,\ell\dot{\theta}\cos\theta$$

Step 2. Rate of change of E_{rel}

Projecting $m\mathbf{a}_{rel} = m\mathbf{g} - T\mathbf{e}_r + \mathbf{F}_{in}$ on $\mathbf{e}_\theta$: $m\ell\ddot{\theta} = -mg\sin\theta - m\ddot{x}_A\cos\theta$. Differentiating E_{rel} with the chain rule, $\frac{d}{dt}\cos\theta = -\sin\theta\,\dot{\theta}$:

$$\dot{E}_{rel} = m\ell\dot{\theta}\left(\ell\ddot{\theta} + g\sin\theta\right) = -m\ddot{x}_A\,\ell\dot{\theta}\cos\theta$$

With $\ddot{x}_A = -b\Omega^2\sin(\Omega t)$: $\dot{E}_{rel} = mb\Omega^2\ell\,\sin(\Omega t)\,\dot{\theta}\cos\theta$. For example at $\Omega t = \pi/2$ with $\theta = 0$ and $\dot{\theta} \neq 0$ it equals $mb\Omega^2\ell\dot{\theta} \neq 0$. E_{rel} is not a first integral: the statement is FALSE.

Step 3. Why no other energy works either

For a constant $\ddot{x}_A = a_0$ (as in a uniformly accelerating support) the inertia force would be uniform and conservative, with potential $ma_0\ell\sin\theta$. Here, if we try $E^* = E_{rel} + m\ddot{x}_A(t)\,\ell\sin\theta$, the explicit time dependence leaves

$$\dot{E}^* = m\,\frac{d^3x_A}{dt^3}\,\ell\sin\theta = -mb\Omega^3\cos(\Omega t)\,\ell\sin\theta \neq 0$$

The motor keeps pumping energy in and out of the bob.

KEY POINTS

- FALSE.
- In S_1: $\mathbf{v}_{rel} = \ell\dot{\theta}\,\mathbf{e}_\theta$, $-T\mathbf{e}_r \cdot \mathbf{v}_{rel} = 0$ (this part is true).
- Equation: $\ell\ddot{\theta} = -g\sin\theta - \ddot{x}_A\cos\theta$; $\dot{E}_{rel} = -m\ddot{x}_A\,\ell\dot{\theta}\cos\theta = mb\Omega^2\ell\sin(\Omega t)\,\dot{\theta}\cos\theta \neq 0$.
- The inertia force depends explicitly on t: even $E_{rel} + m\ddot{x}_A\ell\sin\theta$ changes, at the rate $m\,\dfrac{d^3x_A}{dt^3}\,\ell\sin\theta$.

Interpretation and checks

- **Dimensional check:** $T\dot{x}_A$ is N m s^{-1} = W; $mb\Omega^2\ell\dot{\theta}$ is kg m s^{-2} m s^{-1} = W; $m^2g\ell/(M+m)$ is J.
- **Limits:** $M \to \infty$ makes the slider a fixed support ($\dot{x}_A \to 0$): the energy transfer $m^2g\ell(1-\cos\theta_0)/(M+m) \to 0$ and E_P is conserved. $b \to 0$ in (b) gives the ordinary pendulum and $\dot{E}_{rel} \to 0$.
- **Common mistakes in the exam:** (1) "the tension is perpendicular to the velocity, so it does no work": true only for the velocity relative to the point the string is tied to; (2) forgetting the inertia force when writing the energy in a non-inertial frame; (3) treating a time-dependent inertia force as conservative.

`Conceptual question · Unit 4 · 2 points · 15 min · Conceptual / fast`

C04 — Angular momentum under a non-central force and about a moving point

A point particle P of mass m moves with respect to the inertial frame $S_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$, with no gravity. $r = |\mathbf{r}_P|$ is its distance to O, $\mathbf{u}_r = \mathbf{r}_P / r$, and $\{\mathbf{e}_R, \mathbf{e}_\varphi, \mathbf{k}\}$ is the cylindrical basis about the z axis. The angular momentum about a point A is $\mathbf{H}_A = \mathbf{r}_{AP} \times m\mathbf{v}_P$ (absolute velocity, Notes Eq. 4.48). Decide whether each statement is TRUE or FALSE and justify the answer. A verdict without a justification scores nothing.

Part (a) · 1 point

A force with an azimuthal part

P moves under $\mathbf{F} = f(r)\,\mathbf{u}_r + q(r)\,\mathbf{e}_\varphi$, with f and q known functions and $q \neq 0$. At $t = 0$, P is in the plane $z = 0$ and its velocity is parallel to that plane. Statement: "Since $\mathbf{F}$ is not central, $\mathbf{H}_O$ is not conserved, and therefore the trajectory of P cannot be planar."

Step 1. The plane $z = 0$ is invariant

In Cartesian components $\mathbf{u}_r = (x, y, z)/r$ and $\mathbf{e}_\varphi = (-y, x, 0)/\sqrt{x^2 + y^2}$, so the z component of Newton's law is

$$m\ddot{z} = f(r)\,\frac{z}{r}$$

With $z(0) = 0$ and $\dot{z}(0) = 0$, the function $z(t) \equiv 0$ satisfies this equation and the initial conditions; by uniqueness it is the solution. P never leaves the plane $z = 0$: the trajectory **is** planar.

Step 2. Angular momentum is not conserved

On the plane, $\mathbf{r}_P = R\,\mathbf{e}_R$ (with $R = r$) and $\mathbf{M}_O = \mathbf{r}_P \times \mathbf{F} = R\,\mathbf{e}_R \times \big(f\,\mathbf{e}_R + q\,\mathbf{e}_\varphi\big) = R\,q(R)\,\mathbf{k} \neq \mathbf{0}$. By Notes Eq. 4.53 ($O$ fixed), $\dot{\mathbf{H}}_O = \mathbf{M}_O$. With $\mathbf{v}_P = \dot{R}\,\mathbf{e}_R + R\dot{\varphi}\,\mathbf{e}_\varphi$:

$$\mathbf{H}_O = R\,\mathbf{e}_R \times m\big(\dot{R}\,\mathbf{e}_R + R\dot{\varphi}\,\mathbf{e}_\varphi\big) = mR^2\dot{\varphi}\,\mathbf{k}, \qquad \frac{d}{dt}\big(mR^2\dot{\varphi}\big) = m\big(2R\dot{R}\dot{\varphi} + R^2\ddot{\varphi}\big) = mR\,a_\varphi = R\,q(R)$$

So $\mathbf{H}_O$ is not conserved (the areal velocity $\tfrac{1}{2}R^2\dot{\varphi}$ is not constant either), but it always points along $\mathbf{k}$.

Step 3. Where the reasoning breaks

Conservation of $\mathbf{H}_O$ is sufficient for planar motion (Notes §4.8.1, property 2) but not necessary. Here $\mathbf{M}_O$ is parallel to $\mathbf{H}_O$: only the magnitude changes, and $\mathbf{r}_P \cdot \mathbf{H}_O = 0$ keeps P in the fixed plane normal to $\mathbf{k}$. The statement is FALSE.

KEY POINTS

- FALSE (the first half is true, the conclusion is not).
- $F_z = f(r)\, z/r$: zero on $z = 0$, so with $z(0) = \dot{z}(0) = 0$ the solution keeps $z \equiv 0$: the motion is planar.
- $\mathbf{M}_O = r\, q(r)\, \mathbf{k} \neq \mathbf{0}$ on the plane: $\mathbf{H}_O = mR^2\dot{\varphi}\, \mathbf{k}$ changes magnitude, $\dfrac{d}{dt}\left(mR^2\dot{\varphi}\right) = R\, q(R)$, but keeps its direction $\mathbf{k}$.

Part (b) · 1 point

Angular momentum about a moving point

Now P moves under a central force towards O on a circle of radius ρ with constant angular rate n: $\mathbf{r}_P = \rho(\cos nt\, \mathbf{i} + \sin nt\, \mathbf{j})$. A point A moves with constant velocity, $\mathbf{r}_A = ut\, \mathbf{i}$ ($u > 0$). Statement: "A frame translating with A at constant velocity is inertial, so the angular momentum of P about A, $\mathbf{H}_A$, is conserved just like $\mathbf{H}_O$."

Step 1. The theorem about a moving point

Differentiating $\mathbf{H}_A = (\mathbf{r}_P - \mathbf{r}_A) \times m\mathbf{v}_P$ with the product rule:

$$\frac{d\mathbf{H}_A}{dt} = (\mathbf{v}_P - \mathbf{v}_A) \times m\mathbf{v}_P + (\mathbf{r}_P - \mathbf{r}_A) \times m\mathbf{a}_P = \mathbf{M}_A - \mathbf{v}_A \times m\mathbf{v}_P$$

because $\mathbf{v}_P \times \mathbf{v}_P = \mathbf{0}$ and $m\mathbf{a}_P = \mathbf{F}$ (Notes Eq. 4.52). Both terms must vanish for conservation. The force passes through O, not through A: $\mathbf{M}_A = (\mathbf{r}_P - \mathbf{r}_A) \times \mathbf{F} = -\mathbf{r}_A \times \mathbf{F}$, since $\mathbf{r}_P \times \mathbf{F} = \mathbf{0}$.

Step 2. Direct computation

$\mathbf{v}_P = n\rho(-\sin nt\, \mathbf{i} + \cos nt\, \mathbf{j})$ and $\mathbf{r}_P - \mathbf{r}_A = (\rho\cos nt - ut)\, \mathbf{i} + \rho \sin nt\, \mathbf{j}$, so

$$H_{A,z} = m\left[(\rho\cos nt - ut)\, n\rho\cos nt + \rho\sin nt\; n\rho\sin nt\right] = m\left(n\rho^2 - ut\, n\rho\cos nt\right)$$

Differentiating with the product rule, $\frac{d}{dt}(t\cos nt) = \cos nt - nt\sin nt$:

$$\dot{H}_{A,z} = -mun\rho\left(\cos nt - nt\sin nt\right), \qquad \dot{H}_{A,z}(0) = -mun\rho \neq 0$$

At $t = 0$, $A = O$ and $\mathbf{M}_A = \mathbf{0}$, yet $\mathbf{H}_A$ changes because of $-\mathbf{v}_A \times m\mathbf{v}_P = -u\,\mathbf{i} \times mn\rho\,\mathbf{j} = -mun\rho\,\mathbf{k}$.

Step 3. The inertial frame of A does not help

In the frame S_A that translates with A (inertial, since $\mathbf{a}_A = \mathbf{0}$), Notes Eq. 4.55 gives $\dot{\mathbf{H}}_{A,A} = \mathbf{M}_A$ for $\mathbf{H}_{A,A} = \mathbf{r}_{AP} \times m(\mathbf{v}_P - \mathbf{v}_A)$. With $\mathbf{F} = -mn^2\mathbf{r}_P$, $\mathbf{M}_A = -ut\,\mathbf{i} \times (-mn^2\rho)(\cos nt\,\mathbf{i} + \sin nt\,\mathbf{j}) = mun^2\rho\,t\sin nt\,\mathbf{k}$, nonzero for instance at $nt = \pi/2$. A frame being inertial does not make the force central about its origin. The statement is FALSE.

KEY POINTS

- FALSE.
- $\mathbf{M}_A = \mathbf{r}_{AP} \times \mathbf{F} = -\mathbf{r}_A \times \mathbf{F} \neq \mathbf{0}$ once $A \neq O$, and $\mathbf{v}_A \times m\mathbf{v}_P \neq \mathbf{0}$.
- $H_{A,z} = m\big(n\rho^2 - ut\,n\rho\cos nt\big)$, $\dot{H}_{A,z} = -mun\rho\,(\cos nt - nt\sin nt)$, equal to $-mun\rho \neq 0$ at $t = 0$.
- Even the relative angular momentum $\mathbf{H}_{A,A}$ of Notes Eq. 4.55 changes: $\dot{H}_{A,A,z} = M_{A,z} = mun^2\rho\,t\sin nt$.

Interpretation and checks

- **Dimensional check:** $mR^2\dot{\varphi}$ and $mn\rho^2$ are kg m^2 s^{-1}; $R\,q(R)$ and $mun\rho$ are N m = kg m^2 s^{-2}.
- **Limits:** in (a), $q \to 0$ gives a central force and $mR^2\dot{\varphi}$ constant; if the initial velocity had a z component, $F_z = fz/r$ would not vanish and the motion would in general leave the plane. In (b), $u \to 0$ (A fixed at O) recovers $\dot{\mathbf{H}}_A = \mathbf{0}$.
- **Common mistakes in the exam:** (1) treating "angular momentum conserved" as the only route to planar motion; (2) applying $\dot{\mathbf{H}}_A = \mathbf{M}_A$ about a moving point without the term $-\mathbf{v}_A \times m\mathbf{v}_P$; (3) thinking that a force is central about every point of an inertial frame.

Conceptual question · Unit 5 · 2 points · 15 min · Conceptual / fast

C05 — Work of inertia forces: the Coriolis force on a rotating rod and a free particle in an accelerating frame

> *S_0 is an inertial frame. Decide whether each statement is TRUE or FALSE and justify the answer. A verdict without a justification scores nothing.*

Part (a) · 1 point

Bead on a rotating rod

A straight, smooth rod lies in a horizontal plane and rotates about a fixed vertical axis through its end O with constant angular velocity $\boldsymbol{\Omega} = \Omega\,\mathbf{k}$ ($\mathbf{k}$ upward). A bead of mass m slides on the rod; r is its distance to O, and $S_1 = \{O; \mathbf{e}_r, \mathbf{e}_\varphi, \mathbf{k}\}$ is the frame of the rod ($\mathbf{e}_r$ along the rod). The bead is released at rest relative to the rod at $r = r_0$. Statement: "In S_1 the Coriolis force does work on the bead, and this work is what makes the relative kinetic energy of the bead grow as it slides outwards."

Step 1. Forces in S_1

S_1 rotates with constant $\boldsymbol{\Omega}$ about the fixed point O, so (Notes §4.9) $m\mathbf{a}_{rel} = \mathbf{F} + \mathbf{F}_{cf} + \mathbf{F}_{cor}$ with the centrifugal force $\mathbf{F}_{cf} = -m\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times r\,\mathbf{e}_r) = m\Omega^2 r\,\mathbf{e}_r$ and the Coriolis force

$$\mathbf{F}_{cor} = -2m\,\boldsymbol{\Omega} \times \mathbf{v}_{rel} = -2m\,\Omega\,\mathbf{k} \times \dot r\,\mathbf{e}_r = -2m\Omega\dot r\,\mathbf{e}_\varphi$$

(no Euler term since $\dot{\boldsymbol{\Omega}} = \mathbf{0}$). The real forces are the weight and the rod reaction, both orthogonal to $\mathbf{e}_r$ on a smooth rod.

Step 2. Powers

With $\mathbf{v}_{rel} = \dot r\,\mathbf{e}_r$: $\mathbf{F}_{cor} \cdot \mathbf{v}_{rel} = -2m\Omega\dot r^2\,(\mathbf{e}_\varphi \cdot \mathbf{e}_r) = 0$. This holds for any motion, because $\boldsymbol{\Omega} \times \mathbf{v}_{rel}$ is perpendicular to $\mathbf{v}_{rel}$: **the Coriolis force never works in the moving frame.** Only the centrifugal force works, with power $m\Omega^2 r\dot r$.

Step 3. Relative kinetic energy

Along $\mathbf{e}_r$: $m\ddot r = m\Omega^2 r$, whose solution with $r(0) = r_0$, $\dot r(0) = 0$ is $r = r_0\cosh\Omega t$. Work-energy theorem in S_1, with the work of the centrifugal force from r_0 to r given by Barrow's rule:

$$\tfrac12 m\dot r^2 - 0 = \int_{r_0}^{r} m\Omega^2 s\,ds = \Big[\tfrac12 m\Omega^2 s^2\Big]_{r_0}^{r} = \tfrac12 m\Omega^2\big(r^2 - r_0^2\big)$$

The growth comes from the centrifugal force; the Coriolis force only presses the bead against the rod, where the reaction $N\mathbf{e}_\varphi + N_z\mathbf{k}$ balances it: along $\mathbf{e}_\varphi$, $N = 2m\Omega\dot r$ against the Coriolis force, and along $\mathbf{k}$, $N_z = mg$ against the weight $-mg\,\mathbf{k}$ (both components are orthogonal to the smooth rod). The statement

is FALSE.

Step 4. Seen from S_0

In S_0 there are no inertia forces; the energy is supplied by the rod reaction $N\mathbf{e}_\varphi + mg\,\mathbf{k}$, whose power with the absolute velocity $\mathbf{v} = \dot r\,\mathbf{e}_r + \Omega r\,\mathbf{e}_\varphi$ is $N\Omega r = 2m\Omega^2 r\dot r$ (the vertical part $mg\,\mathbf{k}$ is orthogonal to $\mathbf{v}$), exactly $\frac{d}{dt}\left[\frac{1}{2}m(\dot r^2 + \Omega^2 r^2)\right]$ with $\ddot r = \Omega^2 r$.

KEY POINTS

- FALSE.
- Coriolis force $-2m\boldsymbol{\Omega}\times\mathbf{v}_{rel} = -2m\Omega\dot r\,\mathbf{e}_\varphi \perp \mathbf{v}_{rel}$: zero power, always.
- The work comes from the centrifugal force $m\Omega^2 r\,\mathbf{e}_r$: $\frac{1}{2}m\dot r^2 = \frac{1}{2}m\Omega^2(r^2 - r_0^2)$, $r = r_0\cosh\Omega t$.
- The rod reaction $N\mathbf{e}_\varphi + mg\,\mathbf{k}$, $N = 2m\Omega\dot r$: its horizontal part balances the Coriolis force in S_1 and its vertical part the weight; in S_0 it does work at the rate $2m\Omega^2 r\dot r$.

Part (b) · 1 point

Free particle in an accelerating frame

The frame S_1 translates without rotating, with constant acceleration $\mathbf{a}_{O_1} = a_0\,\mathbf{e}$ ($a_0 > 0$, $\mathbf{e}$ a fixed unit vector) with respect to S_0. A particle subject to no real force (no gravity) is observed from S_1. Statement: "The only inertia force in S_1 is $-ma_0\,\mathbf{e}$, and the relative trajectory of the particle is a straight line only when its initial relative velocity is zero or parallel to $\mathbf{e}$; otherwise it is a parabola."

Step 1. Inertia forces

For a frame in pure translation, $\mathbf{a}_{abs} = \mathbf{a}_{O_1} + \mathbf{a}_{rel}$ (Notes §3.2 with $\boldsymbol{\omega} = \mathbf{0}$), so $m\mathbf{a}_{rel} = \mathbf{F} - m\mathbf{a}_{O_1}$: the only inertia force is $-ma_0\,\mathbf{e}$, uniform and constant. With $\mathbf{F} = \mathbf{0}$: $\mathbf{a}_{rel} = -a_0\,\mathbf{e}$.

Step 2. Relative motion

Integrating twice with Barrow's rule from 0 to t: $\mathbf{v}_{rel} = \mathbf{v}_0 - a_0 t\,\mathbf{e}$ and $\mathbf{r}_{rel} = \mathbf{r}_0 + \mathbf{v}_0 t - \frac{1}{2}a_0 t^2\,\mathbf{e}$. The path is straight exactly when $\mathbf{v}_{rel}$ keeps its direction, i.e. when $\mathbf{v}_{rel}\times\mathbf{a}_{rel} = \mathbf{0}$ (zero curvature, Notes §2.5):

$$\mathbf{v}_{rel}\times\mathbf{a}_{rel} = (\mathbf{v}_0 - a_0 t\,\mathbf{e})\times(-a_0\,\mathbf{e}) = -a_0\,\mathbf{v}_0\times\mathbf{e}$$

which vanishes if and only if $\mathbf{v}_0 = \mathbf{0}$ or $\mathbf{v}_0 \parallel \mathbf{e}$.

Step 3. The parabola

With $\mathbf{e} = \mathbf{i}$ and $\mathbf{v}_0 = U\mathbf{j}$: $x = x_0 - \frac{1}{2}a_0 t^2$, $y = y_0 + Ut$. Eliminating $t = (y - y_0)/U$ gives $x - x_0 = -\dfrac{a_0}{2U^2}(y - y_0)^2$, a parabola. Both claims hold: TRUE. (In S_0 the same particle moves in a straight line at constant velocity.)

KEY POINTS

- TRUE.
- $\mathbf{F}_{in} = -m\,\mathbf{a}_{O_1} = -ma_0\,\mathbf{e}$; $\mathbf{r}_{rel} = \mathbf{r}_0 + \mathbf{v}_0 t - \frac{1}{2}a_0 t^2\,\mathbf{e}$.
- $\mathbf{v}_{rel} \times \mathbf{a}_{rel} = -a_0\,\mathbf{v}_0 \times \mathbf{e}$ (constant): the path is straight if and only if $\mathbf{v}_0 \times \mathbf{e} = \mathbf{0}$.
- Example $\mathbf{v}_0 = U\mathbf{j}$, $\mathbf{e} = \mathbf{i}$: $x - x_0 = -\dfrac{a_0}{2U^2}(y - y_0)^2$.

Interpretation and checks

- **Dimensional check:** $m\Omega\dot{r}$ and ma_0 are N; $m\Omega^2 r^2$ is J; $a_0 y^2/U^2$ is m.
- **Limits:** $\Omega \to 0$ in (a) gives $r \equiv r_0$ (no driving force). $a_0 \to 0$ in (b) makes S_1 inertial and every free trajectory straight.
- **Common mistakes in the exam:** (1) attributing work to the Coriolis force (it is always normal to $\mathbf{v}_{rel}$); (2) adding centrifugal or Coriolis forces in a frame that only translates; (3) "a free particle moves in a straight line" in a non-inertial frame.

Conceptual question · Unit 3 · 2 points · 15 min · Conceptual / fast

C06 — Particle on a moving sphere: degrees of freedom, constraint type and the work of a smooth reaction

*A rigid sphere of radius R has its centre C on the vertical axis Oz of the inertial frame $S_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ ($\mathbf{k}$ upward, gravity $-g\,\mathbf{k}$). A motor moves the sphere without rotating it so that $\mathbf{r}_C = h(t)\,\mathbf{k}$ with the prescribed law $h(t) = H + b\sin\Omega t$ (H, b, Ω constants). A point particle P of mass m rests on the **outer** surface of the sphere, which is smooth, and can leave it. Use spherical coordinates about C: polar angle θ from the upward vertical and azimuth φ about Oz, with the basis $\{\mathbf{u}_r, \mathbf{e}_\theta, \mathbf{e}_\varphi\}$, so that $\mathbf{r}_P = h\,\mathbf{k} + R\,\mathbf{u}_r$ while P is in contact. Decide whether each statement is TRUE or FALSE and justify the answer. A verdict without a justification scores nothing.*

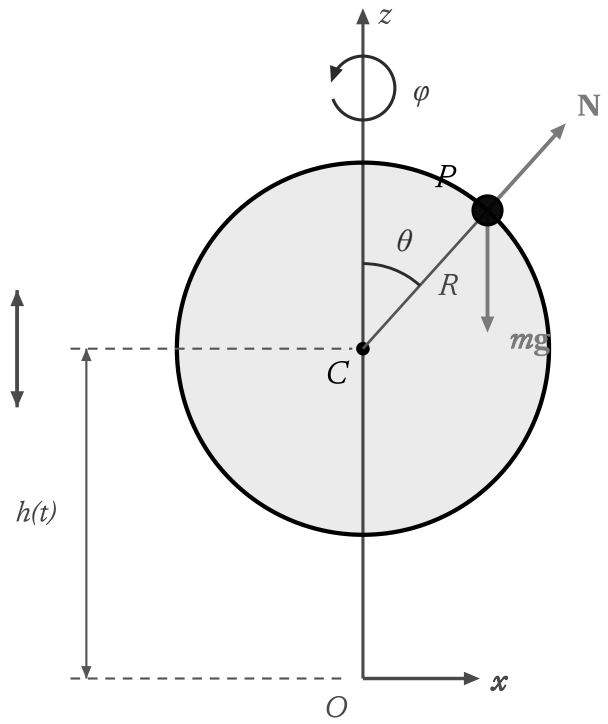


Figure 1. Sphere of radius R with centre C at the prescribed height $h(t) = H + b\sin\Omega t$ on the vertical axis; particle P on its outer surface, polar angle θ from the upward vertical, azimuth φ about the axis.

Part (a) · 1 point

Degrees of freedom

Statement: "While P is in contact with the sphere it has a single degree of freedom, because two conditions are imposed on it: it must lie on the sphere, and the sphere moves with the law $h(t)$."

Step 1. Coordinates and constraint equations

A free particle has 3 degrees of freedom (Notes §2.7). Contact with the sphere imposes one scalar equation on the position of P:

$$f(\mathbf{r},t)=\sqrt{x^2+y^2+\big(z-h(t)\big)^2}-R=0$$

The motion of the sphere, $h(t)$, is a known function of time: it enters this equation but does not add another equation linking x, y, z. It removes no further degree of freedom.

Step 2. Count

$3-1=2$ degrees of freedom; θ and φ fix the position at each instant through $\mathbf{r}_P=h(t)\,\mathbf{k}+R\,\mathbf{u}_r(\theta,\varphi)$. The Lagrangian in (θ,φ) gives two equations of motion, each with its own second derivative (their mass matrix has determinant $m^2R^4\sin^2\theta\neq 0$ off the poles), which confirms two independent coordinates. The statement is FALSE.

KEY POINTS

- FALSE.
- One constraint on P: $f(\mathbf{r},t)=|\mathbf{r}-h(t)\mathbf{k}|-R=0$. The law $h(t)$ is data, not a second constraint.
- $3-1=2$ degrees of freedom while in contact, coordinates (θ,φ); 3 after P leaves the sphere.

Part (b) · 1 point

Classification and work of the reaction

Statement: "The constraint is holonomic, rheonomic and unilateral, and although the sphere is smooth its normal reaction does work on P in S_0."

Step 1. Classification (Notes §2.7)

The constraint $f(\mathbf{r},t)=0$ involves positions and time but no velocities: **holonomic**. It depends explicitly on t through $h(t)$, $\partial f/\partial t=-(z-h)\dot h/|\mathbf{r}-h\mathbf{k}|$, which is not identically zero (it vanishes only where $z=h$ or at the instants with $\dot h=0$): **rheonomic**. P lies on the outside and can lose contact: the true condition is $f\geq 0$ with $N\geq 0$, an inequality: **unilateral**.

Step 2. Velocity of P

Differentiating $\mathbf{r}_P=h\,\mathbf{k}+R\,\mathbf{u}_r$ with $\dot{\mathbf{u}}_r=\dot\theta\,\mathbf{e}_\theta+\sin\theta\,\dot\varphi\,\mathbf{e}_\varphi$ (chain rule on $\mathbf{u}_r(\theta(t),\varphi(t))$):

$$\mathbf{v}_P=\dot h\,\mathbf{k}+R\big(\dot\theta\,\mathbf{e}_\theta+\sin\theta\,\dot\varphi\,\mathbf{e}_\varphi\big),\qquad \dot h=b\Omega\cos\Omega t$$

Step 3. Power of the reaction

Smooth means the reaction is normal to the surface: $\mathbf{N} = N\,\mathbf{u}_r$. With $\mathbf{u}_r \cdot \mathbf{e}_\theta = \mathbf{u}_r \cdot \mathbf{e}_\varphi = 0$ and $\mathbf{u}_r \cdot \mathbf{k} = \cos\theta$:

$$\dot{W}_N = \mathbf{N} \cdot \mathbf{v}_P = N\dot{h}\cos\theta = Nb\Omega\cos(\Omega t)\cos\theta$$

At $t = 0$ with P at the top ($\theta = 0$) it equals $Nb\Omega \neq 0$. A smooth reaction does no work only against a **fixed** surface (Notes §6.4); here the surface pushes P as it moves. In the frame translating with C the velocity of P is $\mathbf{v}_{rel}$, tangent to the sphere, and there $\mathbf{N} \cdot \mathbf{v}_{rel} = 0$. All the claims hold: TRUE.

KEY POINTS

- TRUE.
- Holonomic: f depends on position and time only; rheonomic: $\partial f/\partial t$ not identically zero; unilateral: $f \geq 0$ (outer surface, $N \geq 0$, P may leave).
- $\mathbf{v}_P = \dot{h}\,\mathbf{k} + R\big(\dot{\theta}\,\mathbf{e}_\theta + \sin\theta\,\dot{\varphi}\,\mathbf{e}_\varphi\big)$, $\mathbf{N} \cdot \mathbf{v}_P = N\dot{h}\cos\theta = Nb\Omega\cos\Omega t\cos\theta \neq 0$ in general.
- In the frame translating with C the reaction does no work ($\mathbf{u}_r \cdot \mathbf{v}_{rel} = 0$).

Interpretation and checks

- **Dimensional check:** $N\dot{h}$ is N m s^{-1} = W; f is a length.
- **Limits:** $b \to 0$ (fixed sphere) gives a scleronomic constraint and $\dot{W}_N = 0$: mechanical energy is then conserved. If P were a bead threaded on a fixed meridian, a second constraint would leave 1 degree of freedom.
- **Common mistakes in the exam:** (1) counting a prescribed motion as an extra constraint; (2) "smooth implies no work" for a moving surface; (3) calling the constraint bilateral when the particle is only resting on the outside.

Conceptual question · Unit 2 · 2 points · 15 min · Conceptual / fast

C07 — Damped and forced oscillator: energy decay per period and where the amplitude really peaks

A block of mass $m = 2$ kg moves along the x axis of an inertial frame, attached to a spring of stiffness $k = 50$ N/m and to a viscous damper of constant $c = 4$ N s/m. Its equation of motion is $m\ddot{x} + c\dot{x} + kx = F(t)$, and (Notes Eq. 5.23) $\gamma = c/(2m)$, $\omega_0 = \sqrt{k/m}$. Decide whether each statement is TRUE or FALSE and justify the answer. A verdict without a justification scores nothing.

Part (a) · 1 point

Free motion

With $F(t) = 0$. Statement: "The motion is underdamped, and over each period $T = 2\pi/\omega$ of the damped oscillation the mechanical energy $E = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2$ is multiplied by the factor $e^{-2\pi/\sqrt{6}} \approx 0.0769$, the square of the factor by which the amplitude decays."

Step 1. Type of motion

$\gamma = \dfrac{c}{2m} = \dfrac{4}{4} = 1\ \text{s}^{-1}$ and $\omega_0 = \sqrt{50/2} = 5\ \text{s}^{-1}$. The roots of $mp^2 + cp + k = 0$ are $p = -\gamma \pm \sqrt{\gamma^2 - \omega_0^2} = -1 \pm 2\sqrt{6}\,i$: complex, so the motion is **underdamped** (Notes §5.3.1), with $\omega = \sqrt{\omega_0^2 - \gamma^2} = \sqrt{24} = 2\sqrt{6}\ \text{s}^{-1}$ and $T = 2\pi/\omega \approx 1.283$ s.

Step 2. Exact decay over one period

The general solution is $x = e^{-\gamma t}\big(A\cos\omega t + B\sin\omega t\big)$. Since cos and sin have period $T = 2\pi/\omega$, $x(t+T) = e^{-\gamma(t+T)}\big(A\cos\omega t + B\sin\omega t\big) = e^{-\gamma T}x(t)$, and differentiating this identity, $\dot{x}(t+T) = e^{-\gamma T}\dot{x}(t)$. Both terms of E are quadratic:

$$E(t+T) = e^{-2\gamma T}E(t), \qquad \gamma T = \frac{2\pi}{2\sqrt{6}} = \frac{\pi}{\sqrt{6}}$$

Step 3. Numbers

Amplitude factor $e^{-\pi/\sqrt{6}} \approx 0.277$; energy factor $e^{-2\pi/\sqrt{6}} = \big(e^{-\pi/\sqrt{6}}\big)^2 \approx 0.0769$. The energy is dissipated by the damper: $\dot{E} = \dot{x}\,(m\ddot{x} + kx) = -c\dot{x}^2 \le 0$. The statement is TRUE.

KEY POINTS

- TRUE.
- $\gamma = 1\ \text{s}^{-1} < \omega_0 = 5\ \text{s}^{-1}$: underdamped; $\omega = \sqrt{\omega_0^2 - \gamma^2} = 2\sqrt{6} \approx 4.899\ \text{s}^{-1}$, $T \approx 1.283$ s.
- $x(t+T) = e^{-\gamma T}x(t)$ and $\dot{x}(t+T) = e^{-\gamma T}\dot{x}(t)$ exactly, with $\gamma T = \pi/\sqrt{6}$: amplitude factor $e^{-\pi/\sqrt{6}} \approx 0.277$, energy factor $e^{-2\pi/\sqrt{6}} \approx 0.0769$.

Part (b) · 1 point

Forced motion

Now $F(t) = F_1 \cos \omega_1 t$ with $F_1 = 10$ N. Statement: "In the steady state the amplitude is largest when the forcing frequency equals $\omega_0 = 5\ s^{-1}$; there the response lags the force by 90° and its amplitude is 0.5 m."

Step 1. Steady response at $\omega_1 = \omega_0$

After the transient dies out (Notes §5.4), $x = a_1 \cos(\omega_1 t + \alpha_1)$ with a_1 above and $\tan \alpha_1 = -\dfrac{2\gamma\omega_1}{\omega_0^2 - \omega_1^2}$ (Notes Eq. 5.42). At $\omega_1 = \omega_0$ the denominator of $\tan \alpha_1$ vanishes and $\alpha_1 = -\pi/2$: a lag of 90°. The amplitude is

$$a_1(\omega_0) = \frac{F_1/m}{2\gamma\omega_0} = \frac{10/2}{2 \cdot 1 \cdot 5} = 0.5 \text{ m}$$

Step 2. Where the maximum over ω_1 is

Let $s = \omega_1^2$ and $D(s) = (\omega_0^2 - s)^2 + 4\gamma^2 s$. a_1 is largest where D is smallest:

$$\frac{dD}{ds} = -2(\omega_0^2 - s) + 4\gamma^2 = 0 \implies s^* = \omega_0^2 - 2\gamma^2 = 25 - 2 = 23 \text{ s}^{-2}$$

($d^2D/ds^2 = 2 > 0$: a minimum). Substituting, $D(s^*) = 4\gamma^4 + 4\gamma^2(\omega_0^2 - 2\gamma^2) = 4\gamma^2(\omega_0^2 - \gamma^2)$, so

$$a_{1,max} = \frac{F_1/m}{2\gamma\sqrt{\omega_0^2 - \gamma^2}} = \frac{5}{2\sqrt{24}} \approx 0.510 \text{ m} > 0.5 \text{ m}$$

at $\omega_1 = \sqrt{23} \approx 4.796$ s^{-1}, where $\tan \alpha_1 = -\dfrac{2\sqrt{23}}{2} = -\sqrt{23}$, $\alpha_1 \approx -78.2^\circ$.

Step 3. Verdict

The result "maximum at ω_0" of Notes Eq. 5.45 holds when ω_1 is fixed and ω_0 is varied; when the forcing frequency is varied, the peak is shifted below ω_0 by the damping. The statement is FALSE.

KEY POINTS

- FALSE (the phase and the value at $\omega_1 = \omega_0$ are right; the location of the maximum is not).
- At $\omega_1 = \omega_0$: $a_1 = \dfrac{F_1}{2m\gamma\omega_0} = 0.5$ m, $\tan\alpha_1 \to -\infty$, $\alpha_1 = -90^\circ$.
- Maximum over ω_1 at $\omega_1^2 = \omega_0^2 - 2\gamma^2 = 23$ s^{-2} ($\omega_1 \approx 4.796$ s^{-1}): $a_{1,max} = \dfrac{F_1}{2m\gamma\sqrt{\omega_0^2 - \gamma^2}} = \dfrac{5}{2\sqrt{24}} \approx 0.510$ m > 0.5 m, with $\alpha_1 \approx -78.2^\circ$.

Interpretation and checks

- **Dimensional check:** $\gamma, \omega_0, \omega_1$ are s^{-1}; F_1/m is m s^{-2} and $(F_1/m)/(\gamma\omega_0)$ is m.
- **Limits:** for $\gamma \to 0$ the peak returns to ω_0 and $a_{1,max} \to \infty$; for $\gamma \geq \omega_0/\sqrt{2}$ there is no peak at all ($s^* \leq 0$) and a_1 decreases from its static value $F_1/k = 0.2$ m.
- **Common mistakes in the exam:** (1) using the same decay factor for the amplitude and for the energy; (2) placing the amplitude peak at ω_0 for any damping; (3) confusing the 90° phase lag (exactly at ω_0) with the amplitude maximum.

Conceptual question · Unit 4 · 2 points · 15 min · Conceptual / fast

C08 — Apsidal speeds and a perigee impulse: areal velocity, vis-viva and which apse moves

A satellite moves about the Earth (gravitational parameter μ, the Earth's centre F fixed at a focus) on a Kepler ellipse of eccentricity $e = 1/3$ and perigee radius r_p. Π is the perigee and A the apogee; a is the semi-major axis. Decide whether each statement is TRUE or FALSE and justify the answer. A verdict without a justification scores nothing.

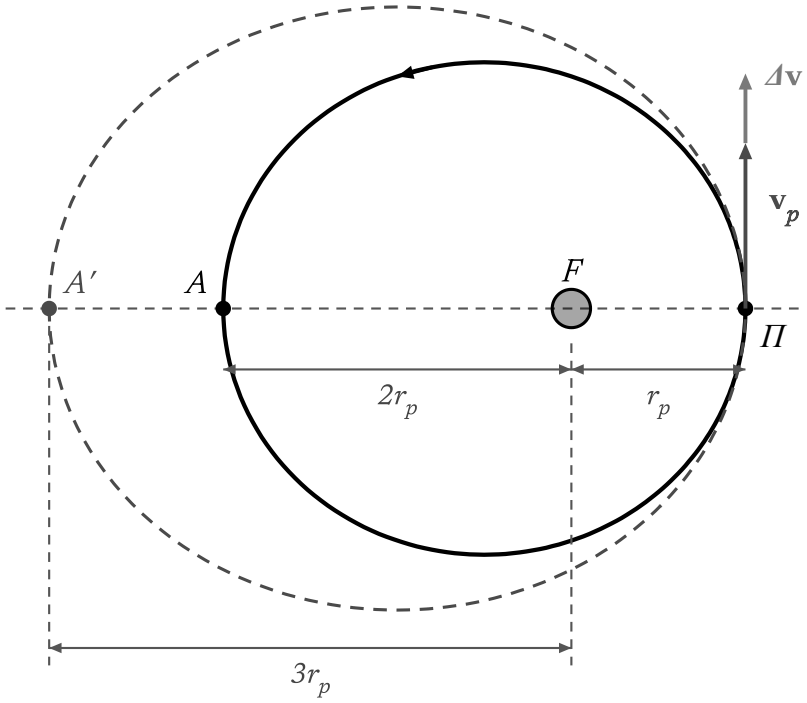


Figure 1. Orbit of eccentricity $e = 1/3$ about the focus F (perigee Π at r_p, apogee A at $r_a = 2r_p$) and, dashed, the orbit after a tangential impulse Δv at Π (apogee A').

Part (a) · 1 point

Speeds and areal velocity

Statement: "The speed at perigee is twice the speed at apogee; consequently the radius vector $\mathbf{r}$ from F to the satellite sweeps area twice as fast when the satellite passes the perigee as when it passes the apogee."

Step 1. Geometry of the ellipse

With $r_p = a(1-e)$ and $r_a = a(1+e)$: $a = \dfrac{r_p}{1-1/3} = \dfrac{3}{2}r_p$ and $r_a = \dfrac{3}{2}r_p \cdot \dfrac{4}{3} = 2r_p$.

Step 2. Speeds at the apsides

The force is central, so $\mathbf{h} = \mathbf{r} \times \mathbf{v}$ is constant (Notes §4.8.1). At both apsides $\mathbf{r} \perp \mathbf{v}$, hence $h = r_p v_p = r_a v_a$ and $\dfrac{v_p}{v_a} = \dfrac{r_a}{r_p} = 2$. The vis-viva equation $v^2 = \mu\left(\frac{2}{r} - \frac{1}{a}\right)$ (Notes §8.5) confirms it:

$$v_p^2 = \mu\left(\frac{2}{r_p} - \frac{2}{3r_p}\right) = \frac{4\mu}{3r_p}, \qquad v_a^2 = \mu\left(\frac{2}{2r_p} - \frac{2}{3r_p}\right) = \frac{\mu}{3r_p}, \qquad \frac{v_p^2}{v_a^2} = 4$$

Step 3. Areal velocity

The area swept per unit time is $\dot{A} = \frac{1}{2}|\mathbf{r} \times \mathbf{v}| = \frac{1}{2}h$, a constant: at perigee $\frac{1}{2} r_p v_p$, at apogee $\frac{1}{2}(2r_p)\,\frac{v_p}{2} = \frac{1}{2} r_p v_p$, the same. The satellite is faster near perigee precisely because it is closer, so that rv stays fixed. The conclusion is wrong: FALSE.

KEY POINTS

- FALSE (the first half is true).
- $r_a = r_p \dfrac{1+e}{1-e} = 2r_p$, $a = \frac{3}{2} r_p$; $h = r_p v_p = r_a v_a \Rightarrow v_p / v_a = 2$. Vis-viva: $v_p^2 = \dfrac{4\mu}{3r_p}$, which equals $\dfrac{2\mu}{a}$ only because $a = \frac{3}{2} r_p$ for this orbit; $v_a^2 = \dfrac{\mu}{3r_p}$.
- Areal velocity $\dot{A} = \frac{1}{2}|\mathbf{r} \times \mathbf{v}| = h/2$: the same at every point of the orbit (Kepler's second law).

Part (b) · 1 point

A tangential impulse at perigee

At Π *a short engine burn along the velocity raises the speed from* v_p *to* $v_p' = \sqrt{9/8}\, v_p$ *(an increase of about* $6.07\,\%$*) without changing the position. Statement: "The perigee radius is unchanged and the apogee radius grows from* $2r_p$ *to* $3r_p$*."*

Step 1. The burn point remains an apse

The impulse changes $\mathbf{v}$ along its own direction, so right after it $\mathbf{v}' \perp \mathbf{r}$ at $r = r_p$: the radial velocity is zero and Π is an apse of the new orbit. Since $v_p'^2 = \dfrac{3\mu}{2r_p}$ exceeds the circular value μ/r_p, the satellite is moving too fast to stay at r_p and rises: Π is the **perigee** of the new orbit.

Step 2. New energy and angular momentum

$$\varepsilon' = \frac{v_p'^2}{2} - \frac{\mu}{r_p} = \frac{3\mu}{4r_p} - \frac{\mu}{r_p} = -\frac{\mu}{4r_p} \;\Rightarrow\; a' = -\frac{\mu}{2\varepsilon'} = 2r_p, \qquad h'^2 = r_p^2 v_p'^2 = \tfrac{3}{2}\mu r_p$$

Step 3. Apse radii

At an apse $v = h'/r$, so $\varepsilon' = \dfrac{h'^2}{2r^2} - \dfrac{\mu}{r}$. Multiplying by r^2: $-\dfrac{\mu}{4r_p}r^2 + \mu r - \dfrac{3}{4}\mu r_p = 0$, i.e. $r^2 - 4r_p r + 3r_p^2 = (r - r_p)(r - 3r_p) = 0$. The perigee stays at r_p and the new apogee A' is at $3r_p$ (consistent with $r_p + r'_a = 2a' = 4r_p$), with $e' = \dfrac{3r_p - r_p}{3r_p + r_p} = \dfrac{1}{2}$. The statement is TRUE.

KEY POINTS

- TRUE.
- $v_p'^2 = \frac{9}{8} \cdot \dfrac{4\mu}{3r_p} = \dfrac{3\mu}{2r_p} > \dfrac{\mu}{r_p}$ (above circular speed, so Π stays the perigee).
- $\varepsilon' = -\dfrac{\mu}{4r_p}$, $a' = 2r_p$; $h'^2 = \frac{3}{2}\mu r_p$; the apse equation $\varepsilon' = \dfrac{h'^2}{2r^2} - \dfrac{\mu}{r}$ has roots $r = r_p$ and $r = 3r_p$; $e' = 1/2$.

Interpretation and checks

- **Dimensional check:** μ is $\mathrm{m^3\ s^{-2}}$, so μ/r is $\mathrm{m^2\ s^{-2}}$ (a speed squared and a specific energy); h^2/μ is a length.
- **Limits:** a burn that brings $v_p'^2$ to $2\mu/r_p$ gives $\varepsilon' = 0$ and an escape parabola ($r'_a \to \infty$); a retrograde burn to $v_p'^2 < \mu/r_p$ would turn Π into the new **apogee**.
- **Common mistakes in the exam:** (1) confusing a constant areal velocity with a constant speed (or the reverse); (2) thinking that a burn raises the point where it is applied, instead of the opposite apse; (3) forgetting that a tangential burn keeps the burn point an apse.